



UNIVERSITÀ DEGLI STUDI DI GENOVA

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DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY,
BIOENGINEERING, ROBOTICS AND SYSTEM ENGINEERING

RESEARCH TRACK 2

Third Assignment

Statistical Analysis

Author:

Amirmahdi Matin

Student ID:

s5884715

Professors:

Carmine Recchiuto

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1 Introduction

A comprehensive statistical analysis is carried out on the first assignment from Research Track 1 as part of the final work of Research Track 2. Simulating and directing a mobile robot in an area was the first task that must be completed for Research Track 1. It was expected that the robot would collect all of the golden boxes that were arranged close to each other in a single location inside the environment. Within the scope of this report, a comparison is made between the performance of my code and the performance of someone else's code that does the same task. Using the hypotheses that are discussed in the next section, the evaluation of performance is carried out. Both the amount of time it takes for the robot to collect all of the golden boxes that are adjacent to one another and the number of times that each algorithm is successful and unsuccessful in detecting and collecting the golden boxes when the number of boxes in the environment changes are the parameters that determine which algorithm performs better.

2 Hypothesis

The statistical method known as hypothesis testing is typically utilized in the field of scientific research in order to arrive at a conclusion regarding the parameters of a data set (such as the mean, variance, and so on).

On the basis of two competing hypotheses, the hypothesis for a particular situation might be described as follows: The null hypothesis (H_0) asserts that the two procedures that are being compared are of identical quality (in accordance with the evaluation methods that will be utilized subsequently). On the other hand, the alternative hypothesis, often known as H_a , is something that is defined as the assertion that one approach is superior to the other method. Both of these hypotheses are false if one of them is true, and vice versa if the other one becomes true. The rejection of the null hypothesis is one way that can be used to demonstrate that the alternative hypothesis is supported by evidence and that one method is superior to the other method. On the basis of this, the null and alternative hypotheses should be defined in the following manner as the initial stage in the process of assessing the two codes contained in this report:

2.1 Null Hypothesis

Both the results produced from my code and the results got from the code of my colleague have the same performance capabilities and are comparable to one another. The average amount of time required to complete this work is the same for both algorithms (i.e., $\mu_A = \mu_B$), which means that we are unable to tell which algorithm is more efficient than the other during the process of gathering the boxes.

2.2 Alternative hypothesis

When it comes to grouping the boxes next to each other under a variety of conditions, the algorithm written by my colleague is far faster than my own approach. Consequently, the average amount of time that my robot takes to collect all of the boxes is longer than the amount of time that my colleague's code takes (i.e., μ_A is greater than μ_B).

2.3 Tests

The first thing that has to be done in order to test the null hypothesis is to collect some sample test data. The amount of data collected throughout the exam ought to be sufficient in order to produce results that are reliable, accurate, and appropriate. The Z-test, the T-test, the two sampled T-test, the paired T-test, and other similar tests are some of the various approaches that can be utilized to examine and compare the performance of two algorithms. In order for us to be able to proceed with the analysis using a certain approach, there are a few fundamental preconditions that must be met. Typically, the Z-test is utilized in situations when the population, as well as its mean and standard deviation, are accessible. When using the T-test, it is necessary to have knowledge about at least one of the characteristics of the population, such as the mean or the standard deviation. For the purpose of this research, the population is not available; rather, we are merely contrasting two distinct algorithms that are capable of completing a task. The paired T-test and the two sampled T-test are both potential usefulnesses for this analysis.

3 Experiment

3.1 Theory

It is necessary to establish identical environments for both algorithms in order to fulfill the need of obtaining test data for the analysis. Each algorithm is put through its paces in a variety of scenarios, with 4, 6, and 8 golden boxes, in order to see how quickly it is able to collect all of the boxes and how accurately it can identify the boxes and collect them. The algorithms are run fifteen times in each environment setup, which results in a total of forty-five simulations for each algorithm. We did not employ random creation of the boxes since we wanted to ensure that each technique was implemented in the exact same setting and that the comparison was accurate. As a consequence of this, the placements of the boxes remain the same in an environment that contains the same quantity of boxes. It has been agreed that a paired T-test would be performed on the data that was extracted. When we simulated both algorithms 45 times with the same amount of randomly generated golden boxes in the environment, we could have also used a two-sample T-test. This would have allowed us to obtain the same results.

The following is the recommended implementation of this method:

- The difference between the two observations for each simulation pair of two algorithms should be calculated using the formula $d_i = y_i - x_i$. It is important to differentiate between positive and negative differences while doing this calculation.
- The mean difference ($\bar{d}$) should be calculated.
- The standard error of the mean difference can be calculated by calculating the standard deviation of the differences, which is denoted by s_d . The formula for calculating the standard error of the mean difference is $SE(\bar{d}) = \frac{s_d}{\sqrt{n}}$.
- Determine the t-statistic, denoted as T, using the formula $T = \frac{\bar{d}}{SE(\bar{d})}$. When the null hypothesis is assumed to be true, this statistic conforms to a t distribution with $n - 1$ degrees of freedom.
- Utilize t-distribution tables to compare your calculated value for T with the t_{n-1} distribution. This will calculate the p-value for the paired t-test.

3.2 Results